

Fully Non-Invasive LIQUID LEVEL DETECTOR

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A non-contact liquid level sensor uses the sensing capacitance of water to detect presence of liquid approaching the sensor. The sensor has a specific static capacitance due to the distributed capacitance.

When the liquid level slowly rises close to the inductor, the parasitic capacitance of the liquid is coupled to the static capacitor, causing the final capacitance of the inductor to become more significant. The capacitance signal is then input to the controller IC for signal conversion. The change in capacitance is converted into a proportional variation of the electrical signal, and then the algorithm detects and judges the degree of the change.

When the amount of change exceeds a certain threshold limit the liquid level is considered to reach the

sensing point. The same is authenticated in the proposed setup through a buzzer (provision for relay and/or LED interface is possible via mode selection).

Further, on the XKC-Y25-NPN sensor a notch is available to change the sensitivity of the sensor. You can rotate the notch clockwise to increase the sensitivity and anti-clockwise to decrease it. No contact with liquid makes the module suitable for hazardous applications, such as detecting toxic substances, strong acids, strong alkali, and all kinds of liquid in an airtight container under high pressure.

If the container is non-metallic, this liquid sensor is easy to use and easy to install. The sound of the buzzer is set at a higher frequency, which catches attention of the operators when the liquid is detected. The waterproof sensor is also compact

and requires very less space to operate. Author's prototype of the proposed setup to detect liquid level in a transparent setup is shown in Fig. 1.

Fig. 2 shows the block diagram representation of the non-invasive liquid level detector. For the present experimental setup, the sensor and components used are listed under Table 1 for the bill of material.

Circuit diagram of the fully non-invasive liquid level detector is shown in Fig. 3. The circuit comprises XKC-Y25 NPN sensor (MOD1) with its inbuilt LED connected to Ar-

TABLE 1
BILL OF MATERIAL

Components	Quantity
XKC-Y25 NPN sensor (MOD1)	1
Arduino Uno (MOD2)	1
Buzzer (B1)	1

TABLE 2
CONNECTIONS BETWEEN XKC-Y25 NPN SENSOR AND THE ARDUINO BOARD

Board Label	Pin Function	Arduino
GND	Ground	GND
5V	5V power input	5V
OUT	Output of the sensor	Pin A2
MODE	Signal selection line	GND
—	Input of buzzer	Pin 8

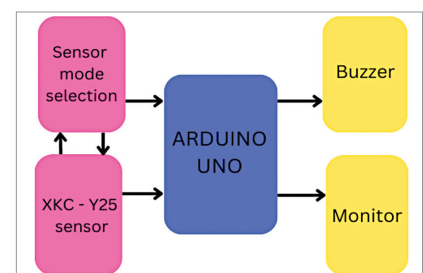


Fig. 2: Block diagram of the non-invasive liquid level detector

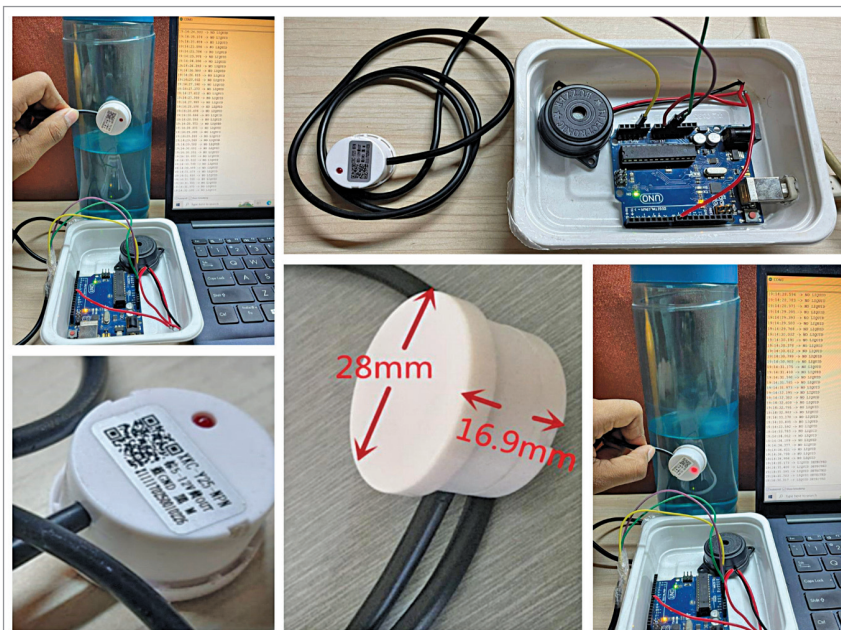


Fig. 1: Author's prototype setup to detect liquid level